

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI-602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SAVEETHA COMPILER ASSESSMENT (LEVEL 1 & VIVA VOCE)

Course Code:	CSA09	Course Title:	Programming in Java
CO Assessed:	CO1 – Manipulation (BL4)	Assessment:	AT1 – Saveetha Compiler (Level 1 & Viva)
Weightage:	20% of CIA	Total Marks:	100 (5 Questions × 20 Marks)
CO1 Statement:	CO1: Apply object-oriented programming principles such as classes, objects, encapsulation, data hiding, inheritance, and polymorphism to design and implement entity manipulations (BL3,BL4)		
Student Name:	P.Sowmya sri	Reg. No.:	192425265
Section / Batch:	SLOT B / 2024	Date:	27/07/2026

Instructions to Students

Answer ALL 5 questions. Total: 100 Marks.

Questions 1–5: Write complete, syntactically correct Java programs on the Saveetha Compiler with proper indentation. Each question carries 20 Marks.

For program questions: test with at least two sample inputs; ensure output is displayed clearly. Each question carries 5 Marks.

No calculators or electronic devices permitted during the viva session.

Question Type	Description	Questions	Marks
Level 1 – Program Writing	Control Flow, classes and Object programs, Collection Framework on compiler	Q1 – Q4	50
GRAND TOTAL:			100 Marks

SECTION A – LEVEL 1: PROGRAM WRITING QUESTIONS (Q1–Q5)

Q1. (Level 1 – Program Writing) [20 Marks]

START

Read Principal
Read Years
Read Choice

IF Choice = 'y' OR Choice = 'Y'
Date = 12

Answer / Program with output:

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JAVA PROGRAMMING

NAGIREDDY HEMANTH REDDY
192472215RB

level 1 Exam
 Level Exam `</> Java` 24 Jul 2026, 02:50 PM

QUESTIONS
 LEVEL 1 — EASY
 Q1 Numbers
 Q2 Loops
 Q3 Strings
 Q4 Array
 Q5 Control
 5 / 5 submitted

Level 1 — Easy (E) 51 — Control
 ✓ Submitted
 Write a program using function to calculate the simple interest. Suppose the customer is a senior citizen. He is being offered 12 percent rate of interest; for all other customers, the ROI is 10 percent.
 Sample Input:
 Enter the principal amount: 200000
 Enter the no of years: 3
 Is customer senior citizen (y/n): n
 Sample Output:
 Interest: 60000
 1. Principal: 2000 , Years: 0
 2. Principal: 20000 , Years: -2
 3. Principal: -2000 , Years: 2
 4. Principal: 2 , Years: 2000
 5. Principal: 0 , Years: 5

Your Code

```

1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         double p = sc.nextDouble();
6         int y = sc.nextInt();
7         char c = sc.next().charAt(0);
8         double r;
9         if (c == 'y' || c == 'Y')
10            r = 12;
11        else
12            r = 10;

```

Output
 Interest: 60000

Q2. (Level 1 – Program Writing) [20 Marks]

START

Declare Matrix A

Declare Matrix B

FOR i = 0 to 1

Answer / Program with output:

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QUESTIONS
 LEVEL 1 — EASY
 Q1 Numbers
 Q2 Loops
 Q3 Strings
 Q4 Array
 Q5 Control
 5 / 5 submitted

Level 1 — Easy (E) Matrix Addition — Array
 ✓ Submitted
 Write a program for matrix addition?
 Sample Input:
 Mat1 =
 1 2
 5 3
 Mat2 =
 2 3
 4 1
 Sample Output:
 Mat Sum =
 3 5
 9 4

Your Code

```

1 class main {
2     public static void main(String[] args) {
3         int a[][] = {{1,2},{5,3}};
4         int b[][] = {{2,3},{4,1}};
5         System.out.println("Mat Sum =");
6         for (int i=0;i<2;i++){
7             for (int j=0;j<2;j++){
8                 System.out.print((a[i][j]+b[i][j])+" ");
9                 System.out.println();
10            }
11        }
12    }

```

Output
 Mat Sum =
 3 5
 9 4

Q3. (Level 1 – Program Writing) [20 Marks]

START

Read Word

Print "Vowels"

FOR each character in Word

IF character is a,e,i,o,u

Print character

Answer / Program with output:



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QUESTIONS
LEVEL 1 — EASY
Q1 Numbers ✓
Q2 Loops ✓
Q3 Strings ✓
Q4 Array ✓
Q5 Control ✓
5 / 5 submitted

Level 1 — Easy (E) Vow's-Con's — Strings ✓ Submitted
Write a program to print consonants and vowels separately in the given word
Sample Input:
Given Word: Engineering
Sample Output:
Consonants: n g n r n g
Vowels: e i e e i
1. TRY
2. MEDIAN
3. ONE
4. KNOWLEDGE
5. EDUCATION
Your Code

```
1 class Main {
2     public static void main(String[] args) {
3         String s = "Engineering";
4         for(int i=0;i<s.length();i++) {
5             char c=s.charAt(i);
6             if(c=='a' || c=='e' || c=='o' || c=='u' ||
7               c=='A' || c=='E' || c=='O' || c=='U')
8                 System.out.print(c+" ");
9         }
10    }
11 }
```

Output

```
E e e
```

Q4. (Level 1 – Program Writing) [20 Marks]

START

Read Character
Read Rows

FOR i = 1 to Rows
FOR j = 1 to i
Print Character

Answer / Program with output:



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QUESTIONS
LEVEL 1 — EASY
Q1 Numbers ✓
Q2 Loops ✓
Q3 Strings ✓
Q4 Array ✓
Q5 Control ✓
5 / 5 submitted

Level 1 — Easy (E) RT-Pattern — Loops ✓ Submitted
Write a program to print the following pattern
Sample Input:
Enter the Character to be printed: %
Max Number of time printed: 3
%
%%
%% %
Your Code

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         char ch = sc.next().charAt(0);
6         int n = sc.nextInt();
7         for (int i=1; i<=n; i++) {
8             for (int j=1; j<=i; j++)
9                 System.out.print(ch + " ");
10            System.out.println();
11        }
12    }
13 }
```

Output

```
%
%%
%% %
```

< Previous

Next >

Q5. (Level 1 – Program Writing) [20 Marks]

START

Read Number

Root = SquareRoot(Number)

Print Root

Print -Root

Answer / Program with output:



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QUESTIONS

LEVEL 1 – EASY

Q1 Numbers ✓

✓

Q2 Loops ✓

✓

Q3 Strings ✓

✓

Q4 Array ✓

✓

Q5 Control ✓

✓

5 / 5 submitted

Level 1 – Easy (E) Sqrt — Numbers

✓ Submitted

Write a program to find the square root of a perfect square number(print both the positive and negative values)

Sample Input:
Enter the number : 6561

Sample Output:
Square Root: 81, -81

1. 1225
2. 9801
3. 1827
4. -100
5. 0

Your Code

```
1 import java.util.Scanner;  
2 public class main {  
3     public static void main(String[] args) {  
4         Scanner sc = new Scanner(System.in);  
5         int n = sc.nextInt();  
6         int root = (int)Math.sqrt(n);  
7         System.out.println(root + ", -" + root);  
8     }  
9 }
```

Output

```
81, -81
```

RUBRICS FOR EVALUATION

Criteria	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (3–4 Marks)	Level 2 – Developing (2 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Analysis & Understanding				
Logic / Algorithm Design				
Python Syntax & String Operations				
Output Correctness & Testing				
Communication & Explanation (Viva)				

Code of Conduct

I p.sowmya sri(192425265) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____